

dCortex Crew Ops Advisor

AI-driven operational
superintelligence for
airline Crew Control.

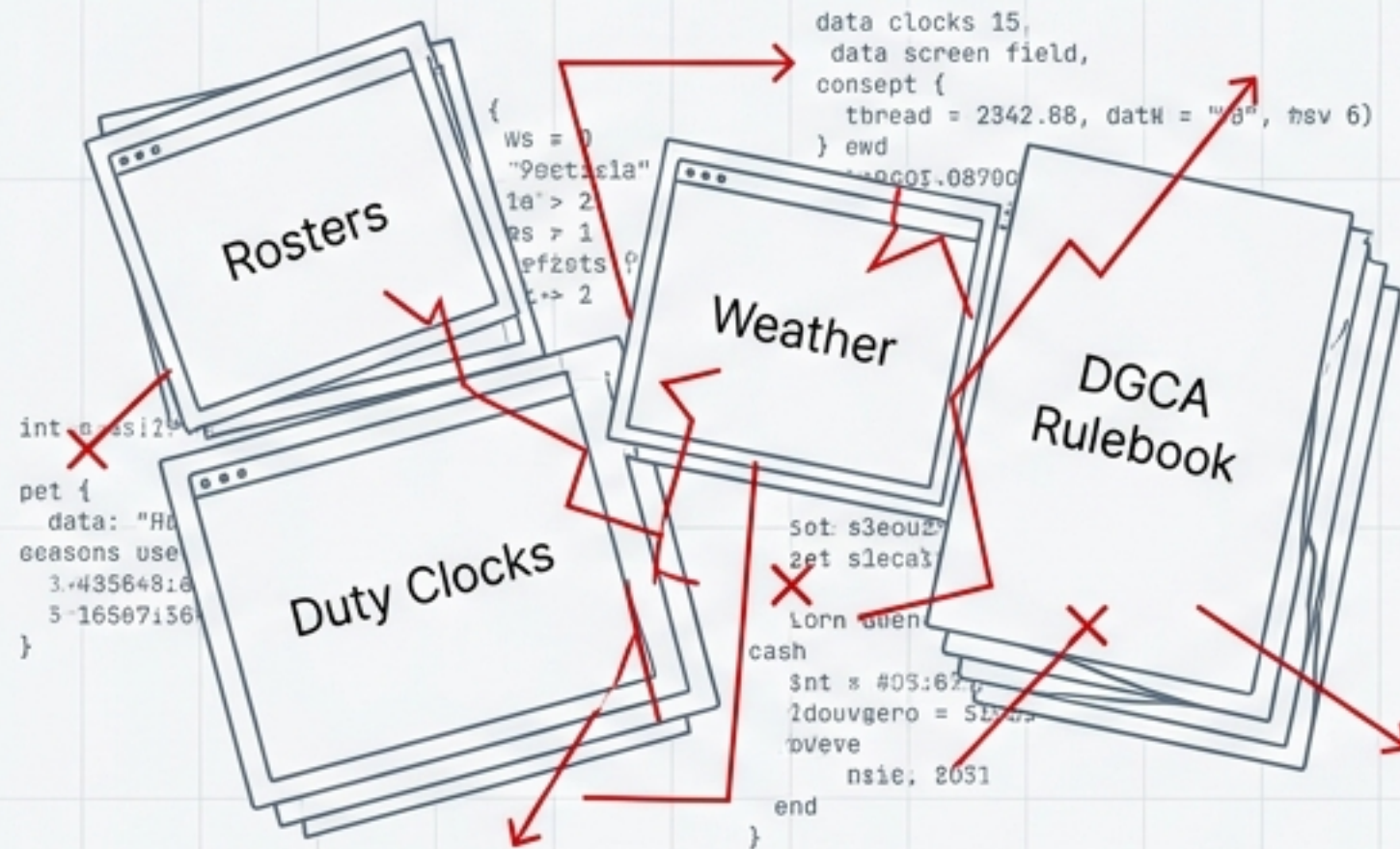
[Crew: C-1042]

[Aircraft: VT-DXC]

[Schedule: P-2291]

The Modern Aviation Blueprint

The Human Bottleneck

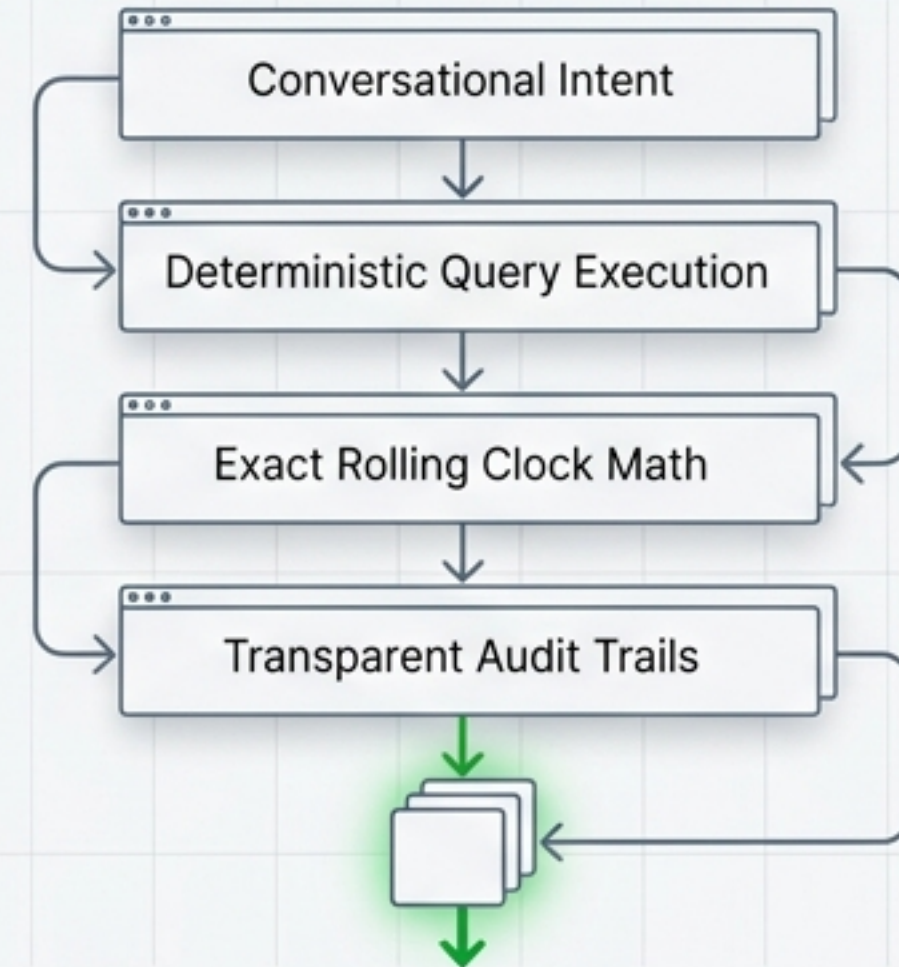


Consequence Blindness

- Fragmented data across screens
- Exact arithmetic under time pressure
- One event cascades invisibly into multi-day downstream breaks

Approximate answers = Regulatory Violations.

The AI Advisor Mandate



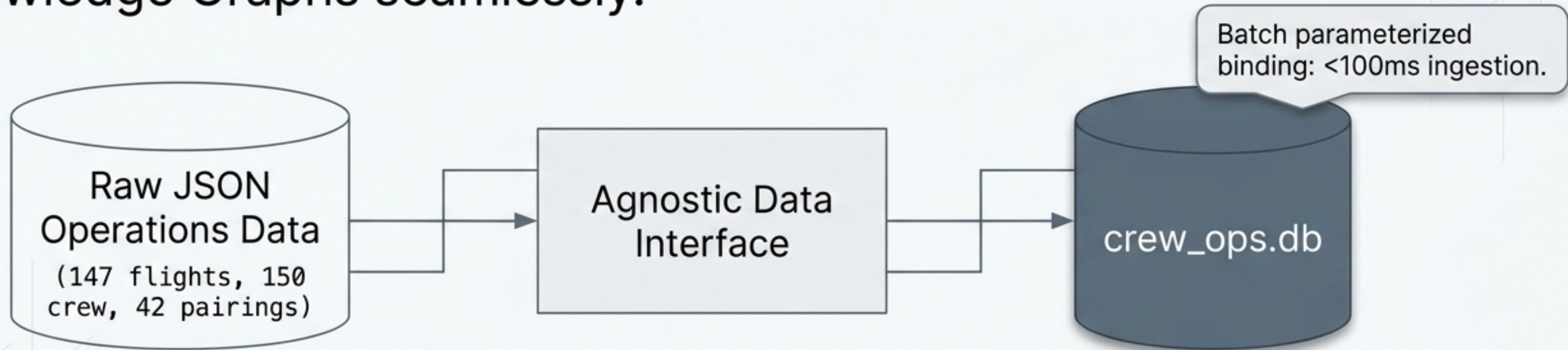
Operational Superintelligence

- Conversational intent mapped to deterministic query execution
- Exact rolling clock math
- Transparent audit trails

Explainable reasoning. Zero hallucinations.

Architectural Philosophy: Data Layer Freedom

The system requires a highly deterministic data store, but **schema decisions are decoupled**. Built to support SQLite, in-memory dicts, or Knowledge Graphs seamlessly.



Note: Rules (`rules.json`) and Financials (`costs.json`) remain in-memory Python constants to ensure immutable domain specification.

The Modern Aviation Blueprint

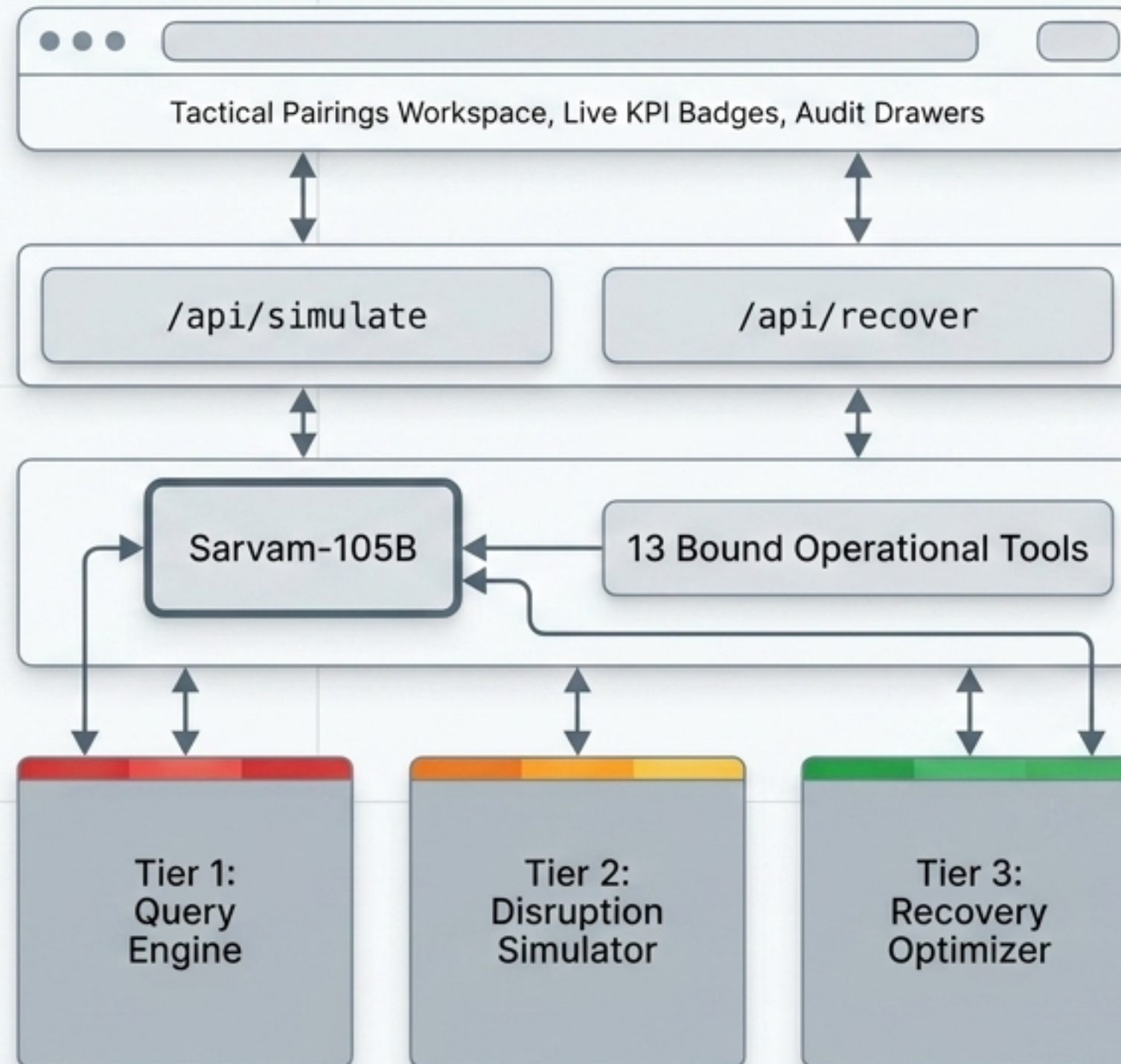
Architectural Diagram: The System Core

React 1:
19 Operations Console

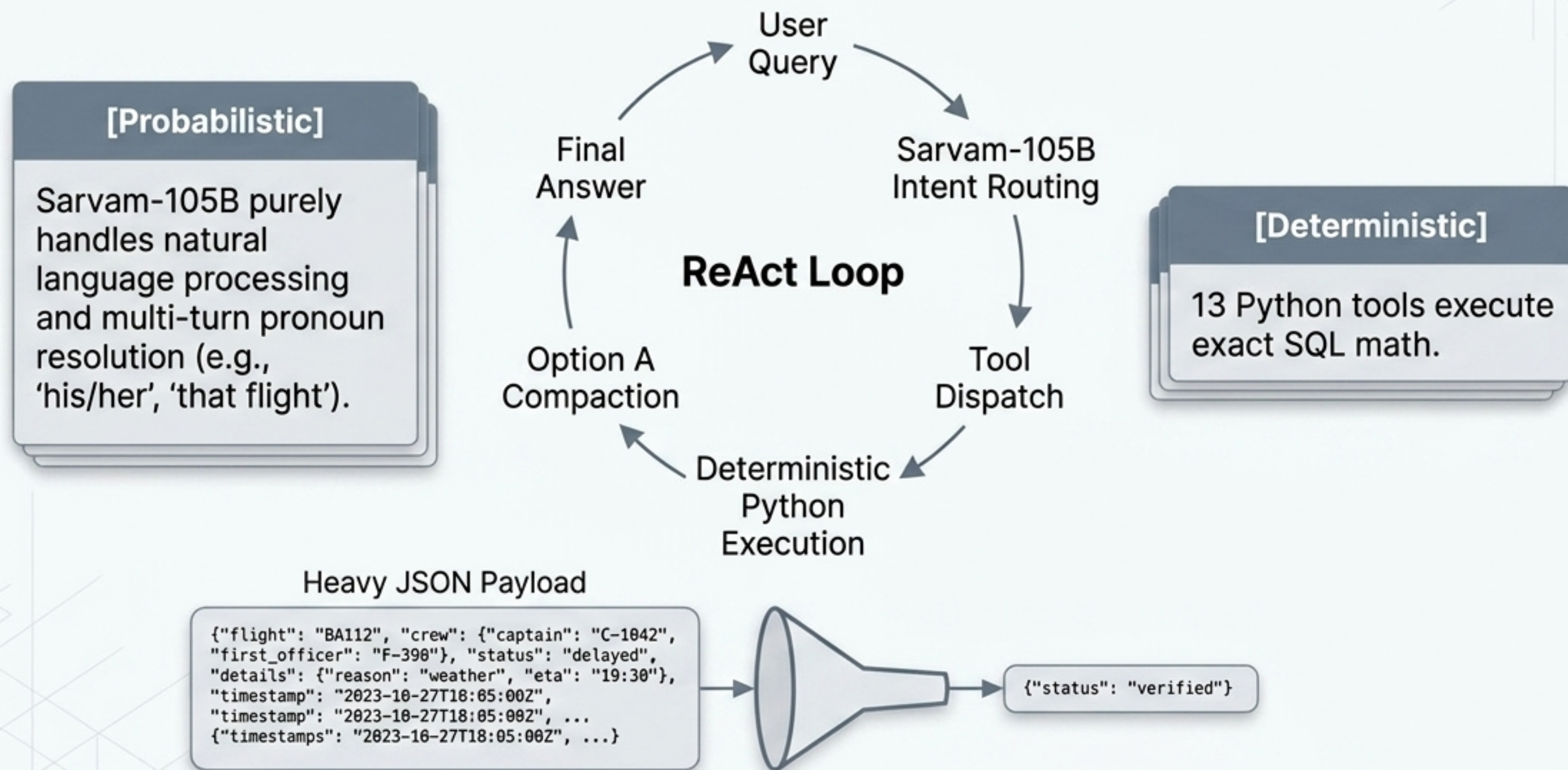
Layer 2:
FastAPI Backend

Layer 3:
LangGraph ReAct Agent (The Brain)

Layer 4:
The Deterministic Engines



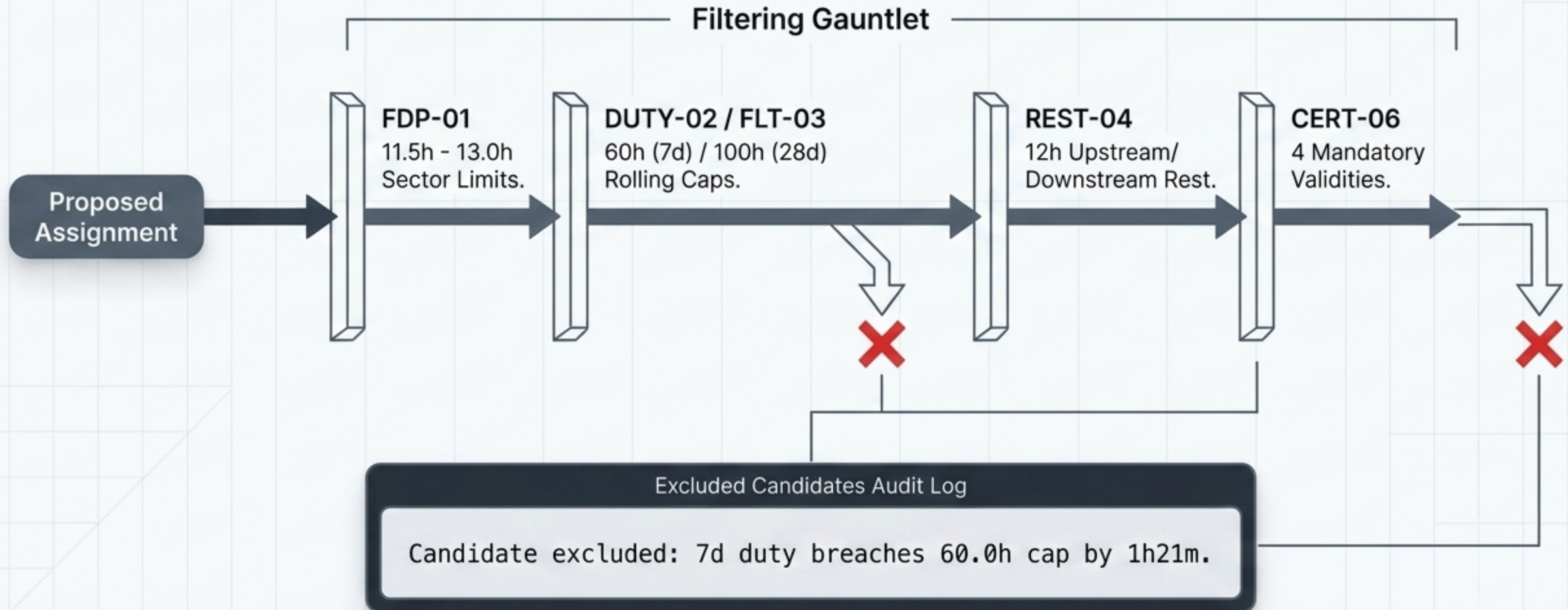
The Modern Aviation Blueprint: ReAct Loop Core



JSON payloads pruned into **verified stubs** to preserve context window and prevent lossy LLM summarization.

The Guardrails: DGCA CAR Legality Engine

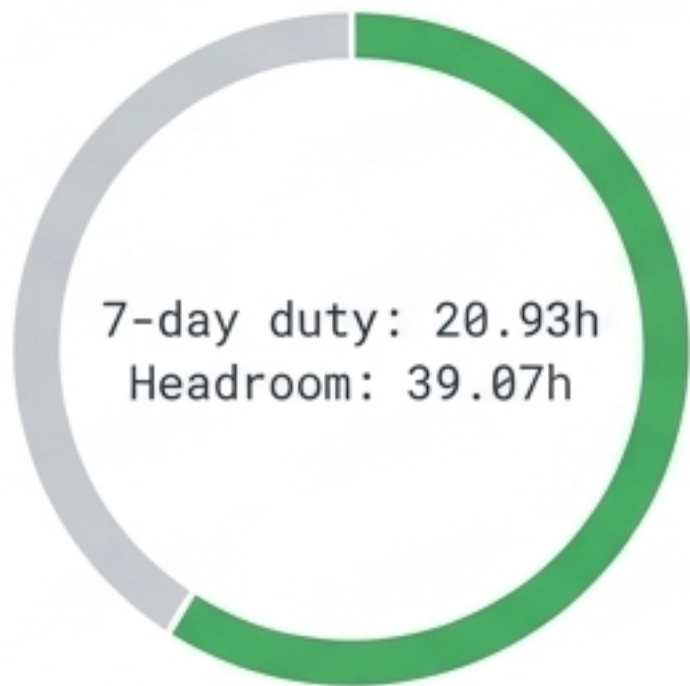
Zero short-circuiting. Every candidate is evaluated against all 7 rules to generate a complete compliance audit.



Tier 1 Insight: Deterministic Lookup & Entity 360

Crew 360

[Crew: C-1042]



Medical (Valid)

Recurrent (Valid)

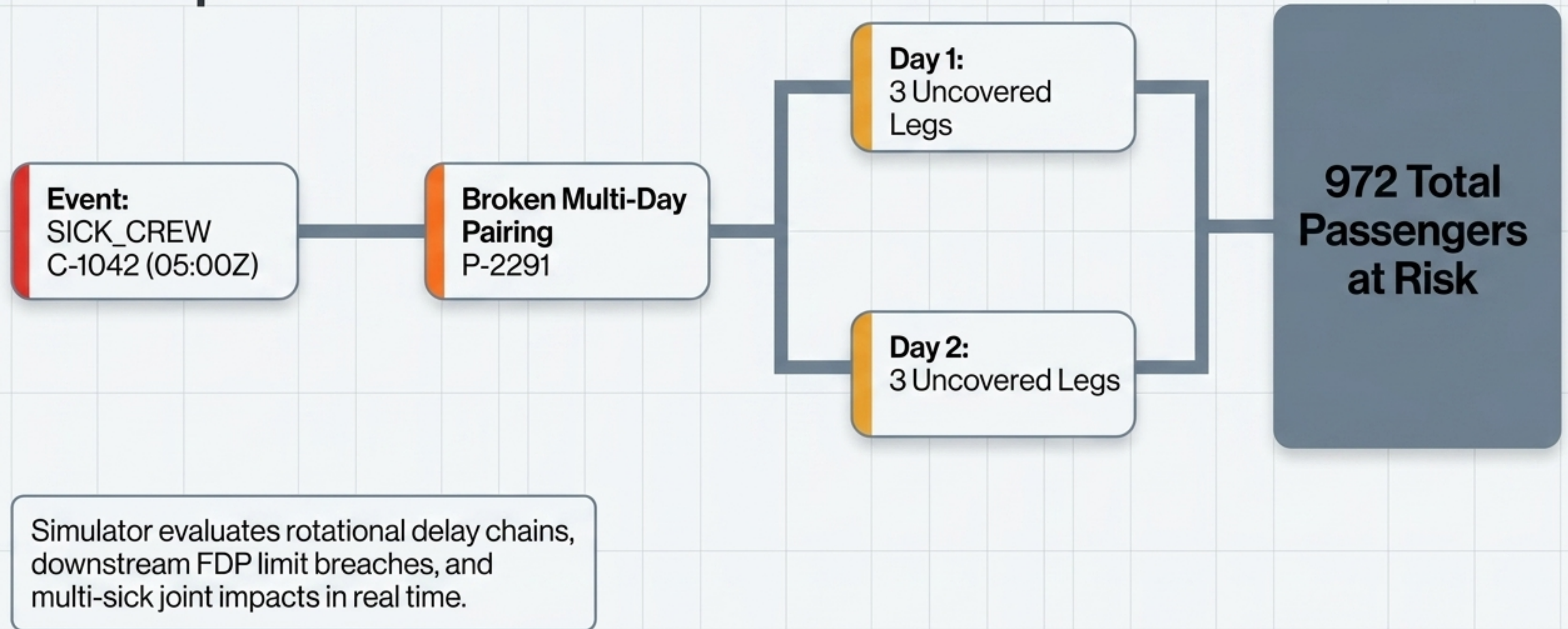
Flight 360

[Flight: DX412]

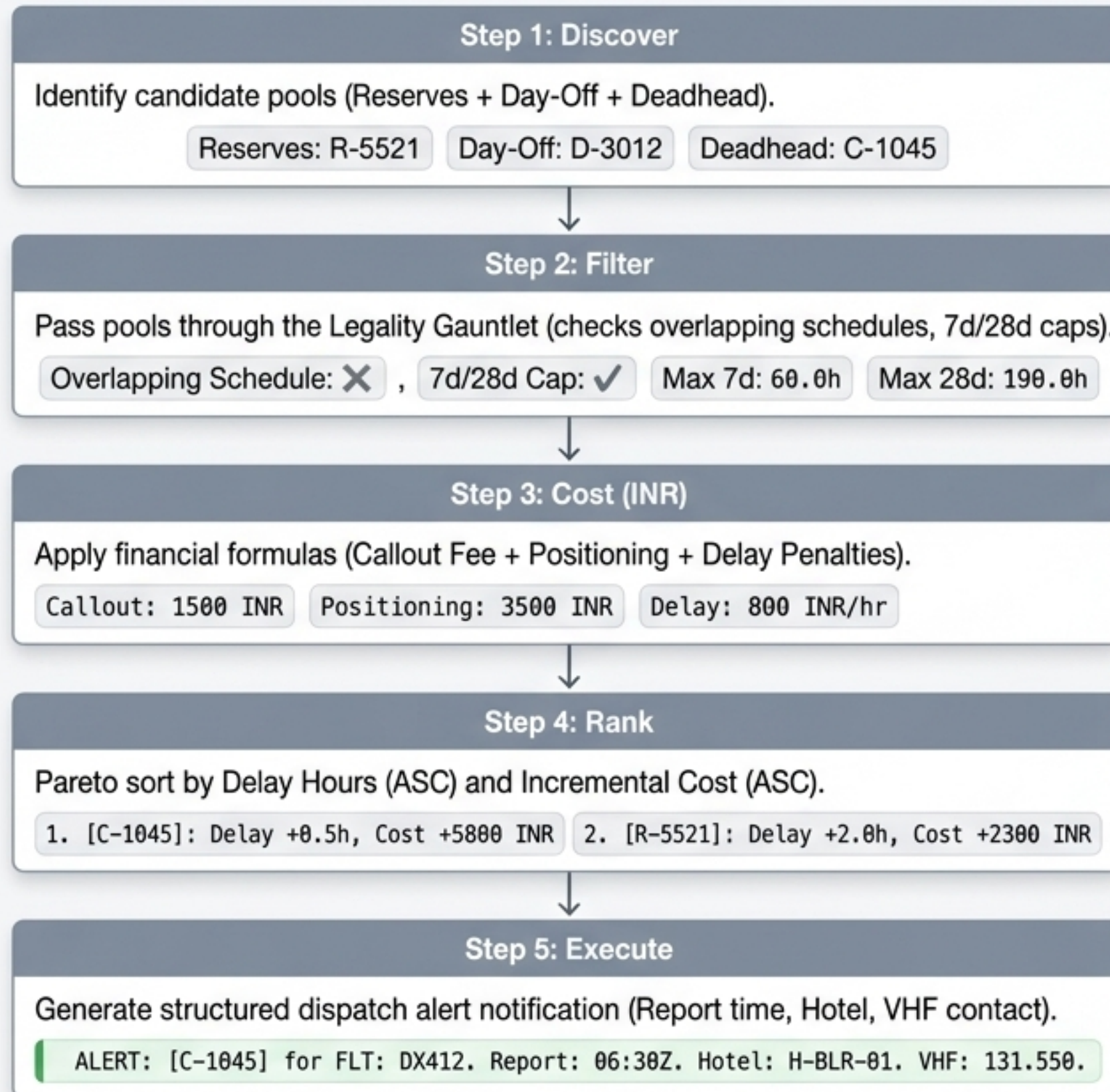


Tail: VT-DXC | Seats: 162 | Block: 2.5h

Tier 2 Intelligence: Disruption Simulation



Tier 3 Optimization: Combinatorial Joint Recovery



In multi-sick scenarios (e.g., Scenario S6), a combinatorial joint solver avoids double-allocating resources, finding the global cost minimum.

Candidate Discovery Matrix

Pool Source	Base Cost (INR)	Delay Impact	Legal Friction
Standby Reserve	₹18,500 (Pilot)	0.0 Hours	Low (Pre-cleared)
Off-Duty (Day-Off)	₹24,000 (Pilot)	0.0 Hours	Med (Requires rest check)
Outstation Deadhead	₹18,500 + ₹6,500 + (Delay*₹5,400)	+2.0 to 4.0 Hours	High (Positioning logistics)
Fallback Cancellation	₹250,000 / flight	N/A	None (Last resort)

The Operations Console: Tactical Workspace

React 19 NOC Operations Workspace

Active Pairings **(42)**

Unassigned Sectors **(0)**

Elevated Risk **(3)**

Crew Complement **(156)**

PAIRING ID	SECTOR	CREW	PAIRING ID	RISK LEVEL	
				[Critical 1.0]	Imminent cert expiry/grounding risk
				[High >0.7]	Active reserve pre-alerted
				[Elevated >0.3]	Near 60h/100h fatigue thresholds
				[Low]	Normal operations

Complete Explainability & Auditability

Reserve Captain C-3315 is the optimal cover at ₹18,500.

Audit Drawer

Tool Dispatched:
`optimize_disruption_recovery(crew="C-3231", event="SICK")`

Rules Checked:
RULE-DUTY-02, RULE-CERT-06...

Excluded Candidate Log:
C-2087 excluded: 7d duty breaches
60.0h cap by 1h21m.

Controllers **never** have to trust a black box. The exact SQLite queries, CAR rule checks, and financial math are fully **exposed**.

Benchmark Matrix & Evaluation

92

Automated Pytest Suite tests passed (APIs, rules, simulators). Execution time: <10 seconds.

38/38 

Operational Benchmark Queries (Q01-Q38) verified against ground-truth JSON data.

6

Core Disruption Scenarios (S1-S6) seamlessly modeled (from sick calls to BLR fog closures).

100% 

DGCA CAR Compliance across evaluated multi-day accumulations and FDP limits.

Includes zero-dependency CLI evaluators for rapid operational auditing.

Synthesis: The Lifecycle of a Disruption

Trigger (0:00)

Event: Weather
Closes BLR (08:00Z -
14:00Z).

[Critical: C-1042]

[Weather: Fog]

Tier 1 (Lookup)

Instantly identifies 13
affected flights and
rostered crew
profiles.



[Flights: 13]

[Crew Profiles: 45]

[Data Source: NOC]

Tier 2 (Simulate)

Calculates reopen
delay + 30m
turnarounds. Flags tail
legs breaching FDP
limits.



[Delay: +4h 30m]

[FDP Breach: 3 Legs]

[Sim Time: 0.2s]

Tier 3 (Optimize)

Discovers legal
Reserve covers for
breached tail legs.
Costs options.



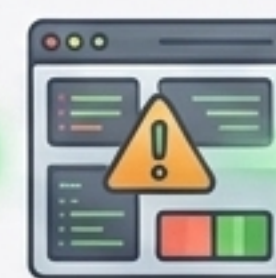
[Legal Covers: 4]

[Cost: ₹18,500]

[Options: 2]

Resolution (0:01)

Generates official
dispatch alerts and
updates risk banding
on the NOC console.



[Alerts Sent: 13]

[Risk Banding: Updated]

[Total Time: 0.01s]

An hours-long manual process is reduced to a single integrated computational second.

From Human Bottleneck to Scalable AI Reasoning

- **Unified Data:** Fragmented silos replaced by a deterministic, sub-millisecond query engine.
- **Exact Math:** Approximate human guesses replaced by rigorous DGCA CAR execution.
- **Consequence Vision:** Multi-day cascade simulations map every exposed passenger seat instantly.

The dCortex Crew Ops Advisor is a fully operational, **benchmark-verified** system of **action**. It does not just chat; it **solves**.